CNN Showcase: Image Classification and Model Explainer

1 What is Machine Learning?

Machine learning is a subfield of artificial intelligence concerned with the development of computational systems that improve their performance on a given task through exposure to data, rather than through explicit, rule-based programming.

2 What is a CNN?

A Convolutional Neural Network (CNN) is a type of deep learning algorithm that is particularly effective for analyzing visual data. It is designed to automatically and adaptively learn spatial hierarchies of features from input images. CNNs are widely used in image classification, object detection, and other computer vision tasks.

3 How does a CNN work?

4 How was this model built

4.1 What is Tensorflow?

TensorFlow is an open-source machine learning framework developed by Google. It provides a comprehensive ecosystem for building and deploying machine learning models, including deep learning architectures like CNNs. TensorFlow supports various platforms and allows for efficient computation on CPUs, GPUs, and TPUs.

4.2 What Dataset was used to train this model

The dataset used to train this model is the "All-Age-Faces Dataset" created by Tsinghua University of Science and Technology which contains a diverse collection of facial images across different age groups. The dataset includes images of men, women, boys, and girls, allowing the model to learn and classify faces based on age and gender.

5 How does this live demo work?

On this live demo, users can upload an image of a face or use the webcam to get a live feed, and the trained CNN model will classify the image into one of the predefined categories, but how does this trained model return new predictions for unseen images?